

Oleksii Krasnoshtanov

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EDUCATION

Breda University of Applied Sciences

Bachelor of Science in Data Science & Artificial Intelligence; 9.1/10 (4.0 GPA)

- BUas Bachelor Scholarship recipient

Breda, Netherlands

Sep 2024 – Present

PROJECTS & TECHNICAL WORK

NPEC Plant Phenotyping System | *Individual Project, BUas*

Nov 2025 – Jan 2026

- Ranked **#2 of 77** in Kaggle competition for plant root segmentation; selected to present results
- Designed autonomous phenotyping system integrating custom PyTorch U-Net segmentation with PyBullet-simulated OT-2 robot; implemented and benchmarked PID and PPO reinforcement learning controllers

Emotion Analysis Pipeline | *Group Project – Dutch Media Client*

Sep – Nov 2025

- Developed production NLP pipeline for a Dutch media client, processing 40-min video episodes end-to-end: Faster-Whisper V3 transcription with VAD, multilingual emotion classification (6 Ekman + neutral), and automated visualisation
- Fine-tuned DeBERTa and Russian transformer models on merged datasets augmented with Mistral-generated synthetic data; cross-validated across languages to stabilise multilingual predictions

CV Pipeline – Plant Phenotyping Platform | *Group Project, BUas*

Feb – Jun 2026

- Developed MLOps platform for plant organ segmentation: U-Net (resnet34 encoder), FastAPI backend, Next.js frontend, PostgreSQL, Prometheus monitoring
- Implemented full ML lifecycle with Airflow DAGs for data versioning and Azure ML training, automated drift detection, feedback-driven retraining, and CI/CD enforcing 80% test coverage
- Deployed to three environments (local Docker Compose, on-premise Portainer, Azure Container Apps) with API-key and session-based authentication

Ukrainian Lecture Transcription Service | *Personal Project*

Jun 2026

- Built self-hosted transcription service for Ukrainian audio on Faster-Whisper (CTranslate2) with automatic GPU selection via NVML; fully offline, no cloud dependencies
- Deployed on 8-GPU university rack via Docker + Portainer with FastAPI backend, live progress bar, and single-page web UI

MARBET Event Support Chatbot | *2-Person Sprint*

Apr – May 2025

- Built self-hosted RAG system (Llama 3.3 70B via Ollama, LangChain, FAISS) for corporate event Q&A with multi-format ingestion and FastAPI backend; client deployed to production

HACKATHONS & COMPETITIONS

BrabantHack 2026 | *Winner – Demcon Deep Tech Track*

Apr 2026

- Won Demcon Deep Tech track: ResNet50 multi-head bounding box regression for shadow-based pedestrian detection (0.626 IoU on held-out set)
- Team of 3; led architecture selection and training pipeline design under 24-hour time constraint

HERE Technologies Hackathon | *RDM Verification Challenge*

2026

- Built XGBoost classification pipeline for Restricted Driving Manoeuvre verification combining probe GPS features with Mapillary and OSM geospatial signals; designed synthetic data scrambling for NDA-compliant delivery

TECHNICAL SKILLS

Languages: Python, SQL

ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, HuggingFace Transformers

NLP & Audio: LangChain, Faster-Whisper, NLTK, FAISS

Computer Vision: OpenCV, U-Net, PyBullet

Infrastructure: Docker, Git, FastAPI, Streamlit, Portainer, Ollama, GitHub Actions CI/CD

MLOps & Cloud: Azure ML, Airflow, Prometheus, MLflow

Data & Analytics: Pandas, NumPy, PostgreSQL, Power BI

Spoken Languages: English (fluent), Ukrainian (native), Russian (fluent)